

CURRENT POSITION

UNIVERSITY OF TURKU

September 2025 – ongoing

Assistant Professor in Robotics and Machine Learning

- My research focuses on enabling robots to grasp and manipulate objects in uncertain and cluttered environments. I combine deep learning, probabilistic reasoning, and tactile perception to build robust manipulation systems.

PREVIOUS POSITIONS

KTH

May 2022 – June 2025

Postdoctoral Researcher in Robotics

- Supervised three doctoral students: Haofei Lu, Zehang Weng, and Alberta Longhini.
- Published 9 peer-reviewed papers accepted to top robotics conferences (CoRL, ICRA, IROS) and journals (RA-L).
- Was involved in the European project [euROBIN](#).

EDUCATION

AALTO UNIVERSITY

October 2017 – March 2022

Ph.D. in Robotics

- My research addressed the problem of probabilistic 6 degree-of-freedom multi-finger grasping of objects in clutter. My novel solution was to explicitly shape-complete each object in the scene using deep learning and plan grasps on those reconstructions. I also focused on using physics simulators to gather synthetic data for training deep networks.

AALTO UNIVERSITY

2014 – 2016

M.Sc. in Space Science and Technology

- I majored in Space Robotics and Automation. GPA: 4.03/5.

AALTO UNIVERSITY

2011 – 2014

B.Sc. in Automation and Systems Technology

- I majored in Automation and Control Engineering and minored in discrete mathematics. GPA: 4.04/5.

PUBLICATIONS

Journal Papers

- Patrizio Perugini, **Jens Lundell**, Katharina Friedl, and Danica Kragic. "Pushing Everything Everywhere All At Once: Probabilistic Prehensile Pushing". In: *IEEE Robotics and Automation Letters* (2025).
- Zehang Weng, Haofei Lu, Danica Kragic, and **Jens Lundell**. "Dexdiffuser: Generating dexterous grasps with diffusion models". In: *IEEE Robotics and Automation Letters* (2024).
- Zehang Weng, Haofei Lu, **Jens Lundell**, and Danica Kragic. "CAPGrasp: An $\mathbb{R}^3 \times \text{SO}(2)$ -Equivariant Continuous Approach-Constrained Generative Grasp Sampler". In: *IEEE Robotics and Automation Letters* 9.4 (2024), pp. 3641–3647.
- Tran Nguyen Le, **Jens Lundell**, Fares J Abu-Dakka, and Ville Kyrki. "Deformation-aware data-driven grasp synthesis". In: *IEEE Robotics and Automation Letters* 7.2 (2022), pp. 3038–3045.
- Joni Pajarinen, **Jens Lundell**, and Ville Kyrki. "POMDP planning under object composition uncertainty: Application to robotic manipulation". In: *IEEE Transactions on Robotics* 39.1 (2022), pp. 41–56.
- Lukas Rustler, **Jens Lundell**, Jan Kristof Behrens, Ville Kyrki, and Matej Hoffmann. "Active visuo-haptic object shape completion". In: *IEEE Robotics and Automation Letters* 7.2 (2022), pp. 5254–5261.
- **Jens Lundell**, Francesco Verdoja, and Ville Kyrki. "Ddgc: Generative deep dexterous grasping in clutter". In: *IEEE Robotics and Automation Letters* 6.4 (2021), pp. 6899–6906.

Conference Papers

- Katharina Friedl, Noémie Jaquier, **Jens Lundell**, Tamim Asfour, and Danica Kragic. "A Riemannian Framework for Learning Reduced-order Lagrangian Dynamics". In: *The Thirteenth International Conference on Learning Representations*. 2025.
- Daniel Marta, Simon Holk, Miguel Vasco, **Jens Lundell**, Timon Homberger, Finn Busch, Olov Andersson, Danica Kragic, and Iolanda Leite. "FLoRA: Sample-Efficient Preference-based RL via Low-Rank Style Adaptation of Reward Functions". In: *arXiv preprint arXiv:2504.10002* (2025).
- Alberta Longhini, Marcel Büsching, Bardienus Pieter Duisterhof, **Jens Lundell**, Jeffrey Ichnowski, Mårten Björkman, and Danica Kragic. "Cloth-Splatting: 3D State Estimation from RGB Supervision for Deformable Objects". In: *8th Annual Conference on Robot Learning*. 2024.
- **Jens Lundell**, Francesco Verdoja, Tran Nguyen Le, Arsalan Mousavian, Dieter Fox, and Ville Kyrki. "Constrained generative sampling of 6-dof grasps". In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2023, pp. 2940–2946.
- Michael C Welle, Martina Lippi, Haofer Lu, **Jens Lundell**, Andrea Gasparri, and Danica Kragic. "Enabling robot manipulation of soft and rigid objects with vision-based tactile sensors". In: *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)*. IEEE. 2023, pp. 1–7.
- Zehang Weng, Haofer Lu, **Jens Lundell**, and Danica Kragic. "GoNet: An approach-constrained generative grasp sampling network". In: *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*. IEEE. 2023, pp. 1–7.
- Tran Nguyen Le, **Jens Lundell**, Fares J Abu-Dakka, and Ville Kyrki. "A novel simulation-based quality metric for evaluating grasps on 3d deformable objects". In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2022, pp. 3123–3129.
- **Jens Lundell**, Enric Corona, Tran Nguyen Le, Francesco Verdoja, Philippe Weinzaepfel, Grégory Rogez, Francesc Moreno-Noguer, and Ville Kyrki. "Multi-fingan: Generative coarse-to-fine sampling of multi-finger grasps". In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2021, pp. 4495–4501.
- Tran Nguyen Le, **Jens Lundell**, and Ville Kyrki. "Safe grasping with a force controlled soft robotic hand". In: *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE. 2020, pp. 342–349.
- **Jens Lundell**, Francesco Verdoja, and Ville Kyrki. "Beyond top-grasps through scene completion". In: *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2020, pp. 545–551.
- Dennis Ehlers, Markku Suomalainen, **Jens Lundell**, and Ville Kyrki. "Imitating human search strategies for assembly". In: *2019 International Conference on Robotics and Automation (ICRA)*. IEEE. 2019, pp. 7821–7827.
- **Jens Lundell**, Francesco Verdoja, and Ville Kyrki. "Robust grasp planning over uncertain shape completions". In: *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2019, pp. 1526–1532.
- Francesco Verdoja, **Jens Lundell**, and Ville Kyrki. "Deep network uncertainty maps for indoor navigation". In: *2019 IEEE-RAS 19th International Conference on Humanoid Robots (Humanoids)*. IEEE. 2019, pp. 112–119.
- **Jens Lundell**, Francesco Verdoja, and Ville Kyrki. "Hallucinating robots: Inferring obstacle distances from partial laser measurements". In: *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2018, pp. 4781–4787.
- **Jens Lundell**, Murtaza Hazara, and Ville Kyrki. "Generalizing Movement Primitives to New Situations". In: *Conference Towards Autonomous Robotic Systems*. Springer. 2017, pp. 16–31.

Preprints

- Haofei Lu, Yifei Dong, Zehang Weng, Florian Pokorny, **Jens Lundell**, and Danica Kragic. "Grasping a Handful: Sequential Multi-Object Dexterous Grasp Generation". In: *arXiv preprint arXiv:2503.22370* (2025).

OTHER ACADEMIC EXPERIENCES

RESEARCH VISIT

January 2020 – March 2020 / Seattle

University of Washington, State Estimation Lab

- Research visit under the supervision of Professor Dieter Fox. I was working on transferring robotic grasps from one object to another, resulting in the following well-received open-source robotic grasping package https://github.com/jsll/pytorch_6dof-grasping.

CONFERENCE PRESENTATIONS

Humanoids 2018, IROS 2018, IROS 2019, ICRA 2020, IROS 2021

TEACHING EXPERIENCES

TEACHING ASSISTANT

Spring 2019 and 2021 / Helsinki

Aalto University

- Teaching assistant for professor Ville Kyrki's robotic manipulation course. I developed all exercises in ROS and MuJoCo. I also held exercise sessions where I helped students with various homework problems.

MASTER'S THESIS ADVISOR

January 2018 – ongoing

- Helped the master thesis workers define the research problem, gave feedback on writing, and mentored them daily throughout the project. My advising has resulted in 8 accepted master theses, out of which 2 were turned into successful peer-reviewed publications.

GRANTS

AALTO ELEC DOCTORAL SCHOOL SCHOLARSHIP

2019

- Funds covering my salary for 3 years of my doctoral studies.

AWARDS

AALTO UNIVERSITY DOCTORAL THESIS AWARD

2023

- A 3000€ award given to the most meritorious top ten percent of the doctoral theses at the School of Electrical Engineering. 4 such awards were given in 2023.

COLLABORATIONS

CTU PRAGUE

Summer 2021

- I initiated a collaboration on visuo-haptic shape completion with Matej Hoffmann and one of his master's students. This collaboration resulted in a paper that is currently in review for ICRA 2022.

NAVERLABS

Winter 2020

- I and Ville Kyrki initiated a collaboration on multi-finger grasping together with Grégory Rogez from Naver Labs and Francesc Moreno-Noguer from CSIC-UPC. This collaboration resulted in the Multi-FinGAN paper presented at ICRA 2021.

ÖREBRO UNIVERSITY

Spring 2018

- I initiated a collaboration on reinforcement learning together with Todor Stoyanov from Örebro University. This collaboration resulted in the safe-to-explore state-spaces paper presented at Humanoids 2018.

SKILLS

PROGRAMMING LANGUAGES *Experienced:* Python *Familiar:* C++ | C | Julia | R

FRAMEWORKS & LIBRARIES ROS | PyTorch | Tensorflow | MuJoCo | PyBullet | GIT | Isaac Gym

ROBOTS Franka Emika Panda | Kuka LWR 4+ | Care-O-bot 4

LANGUAGES *Native:* Swedish *Fluent:* English *Intermediate:* Finnish

REFERENCES

- Prof. Ville Kyrki, Aalto University, ville.kyrki@aalto.fi
- Prof. Danica Kragic, KTH, dani@kth.se